

1. Remove the JP1 from the Reflow board

2. Prepare Arduino Nano as ISP

Before connecting the ATtiny85, you must configure your Nano to act as a programmer.

1. Open the **Arduino IDE**.
2. Go to **File > Examples > 11.ArduinoISP > ArduinoISP**.
3. Go to **Tools > Board** and select **Arduino Nano**.
4. Go to **Tools > Port** and select your Nano's COM port.
5. Click **Upload** to burn the sketch onto your Nano

3. Wiring Hardware Connections

Disconnect the Arduino Nano from your computer before wiring. Connect the Nano pins to the corresponding SOIC8 pins using your test clip or adapter.

Arduino Nano Pin	Board Pin	Pin Function
D13	Pin 4	SCK (Serial Clock)
D12	Pin 2	MISO (Master In Slave Out)
D11	Pin 3	MOSI (Master Out Slave In)
D10	Pin 6	RESET
5V	Pin 1	VCC (+5V Power)
GND	Pin 5	GND (Ground)

 **Critical Step:** Connect a **10 µF capacitor** between the Nano's **RESET** and **GND** pins. The capacitor's **negative/striped leg goes to GND**, and the **positive leg goes to RESET**. This prevents the Nano from resetting itself while sending code to the ATtiny85.

4. Install ATtiny85 Board Support

You must add ATtiny support to the Arduino IDE to compile code for this chip.

1. Go to **File > Preferences**.
2. Locate the **Additional Boards Manager URLs** field and paste this URL:
`https://raw.githubusercontent.com/damellis/attiny/ide-1.6.x-boards-manager/package\_damellis\_attiny\_index.json`
3. Click **OK**.
4. Go to **Tools > Board > Boards Manager**.
5. Search for `attiny` and click **Install** on the package by **David A. Mellis**.

5. Configure IDE Tools

Plug the Arduino Nano back into your computer and set up the following parameters:

1. Go to **Tools** > **Board** > select **ATtiny25/45/85**.
2. Go to **Tools** > **Processor** > select **ATtiny85**.
3. Go to **Tools** > **Clock** > select **Internal 1 MHz** (or **Internal 8 MHz** if your project requires higher speeds).
4. Go to **Tools** > **Programmer** > select **Arduino as ISP** (Do *not* select "ArduinoISP").

6. Burn Bootloader (Optional but Recommended)

If this is a brand-new ATtiny85 chip, or if you changed the clock speed settings in the previous step, you must configure the internal fuses.

1. Go to **Tools**.
2. Click **Burn Bootloader**.
3. Wait for the confirmation message "Done burning bootloader".

7. Upload

Now you can flash your custom code directly to the surface-mount chip.

1. Open "tinycode3"
2. Go to **Sketch** > Click **Upload Using Programmer** (or press `Ctrl + Shift + U`).

8. Return back the JP1 to Reflow board